Shiftless Collatz Model:

Binary Interference in the 3n+LSB Trajectory

Hiroshi Harada — March 21, 2026

Abstract

The Shiftless Collatz model is a reformulation of the classical Collatz mapping that eliminates the right-shift operation ($n/2$) entirely. Instead of discarding the least significant bit (LSB), the model injects its full binary weight (2^v) into the system. This preserves all bit-level information and reveals internal structures—such as carry avalanches, Sierpiński-like patterns, and moiré-style interference—that remain hidden in the standard Collatz process. This report presents the mathematical equivalence between the Shiftless and standard Collatz models, visualizes the resulting fractal trajectories, and demonstrates a complete linear decomposition of the Shiftless dynamics.

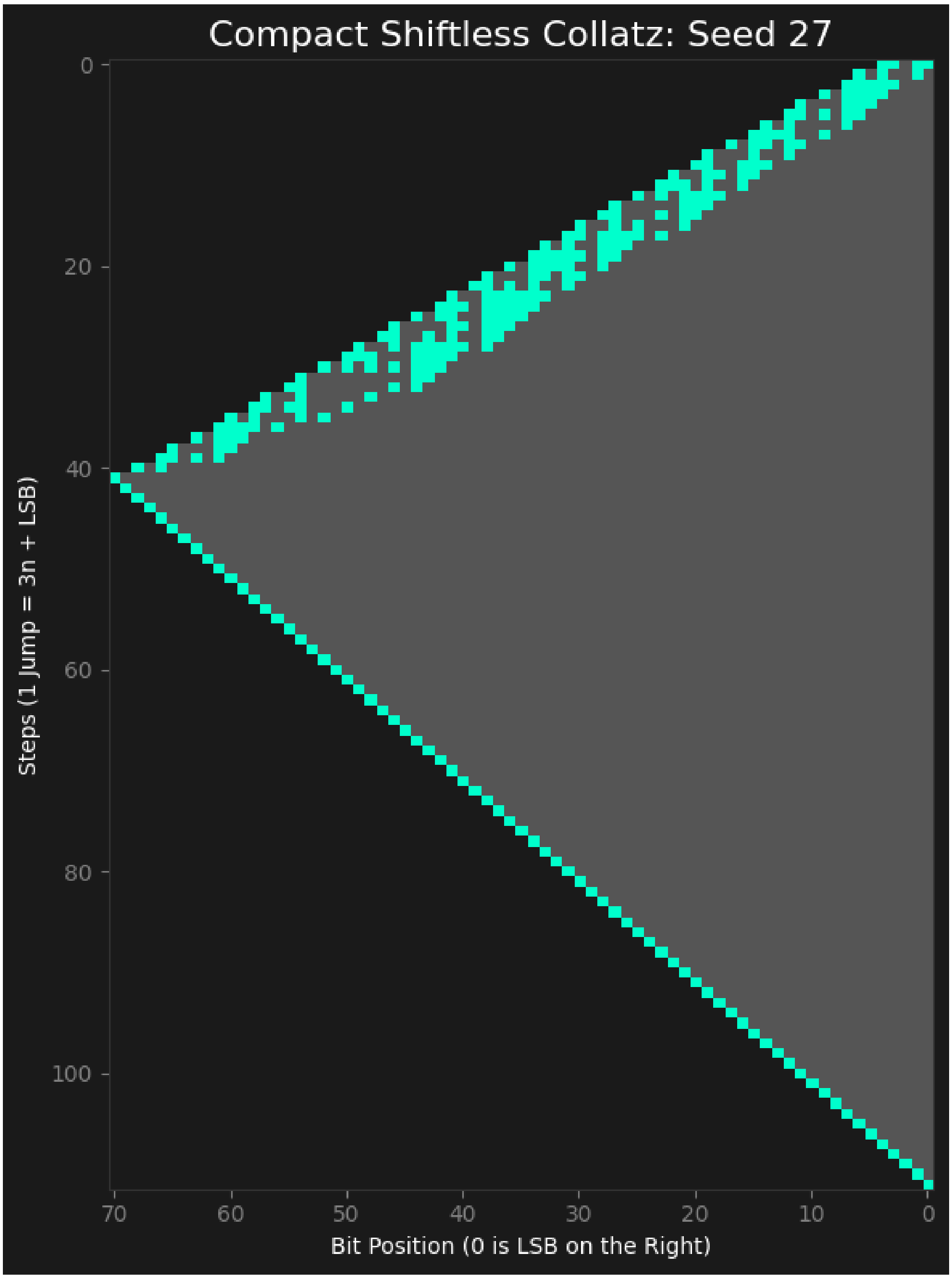

Figure 1. Representative visualization of the Shiftless Collatz trajectory, showing simultaneous emergence of lower-bit carry avalanches and upper-bit Sierpiński-gasket structures.